Gyaansetu – Main Website PRD (v1.0)

*The public front door of the Gyaansetu platform: one scrolling website that opens with the logo and tagline, then walks the visitor through **About Us**, **Our App** and **Our Extension**.*

1. Overview

Item	Detail
Product	Gyaansetu Main Website (the “hub”)
Type	Static, single-page marketing site. No accounts, no backend, no database
Purpose	Explain the platform in one place and send visitors to the working web app and the working extension site
Audience	Students and families, teachers and institutions, hackathon judges, partners and sponsors
Stack	Vite, React, Tailwind CSS; GSAP ScrollTrigger and Lenis for scroll; Framer Motion for cursor and micro-interactions; tsParticles for the background
Hosting	Vercel or Netlify (GitHub Pages possible)
Links	Two external destinations supplied after hosting: the web app URL and the extension website URL

Gyaansetu (ज्ञानसेतु) means *bridge of knowledge*: gyaan = knowledge, setu = bridge. The platform has three parts that this site must keep distinct and connected:

1. **This main website**: explains who we are and why.
2. **The web app**: sign-up, onboarding, dashboard, saved preferences and connected devices.
3. **The extension**: Chrome and Edge extension that applies the saved profile on YouTube, Physics Wallah and Unacademy.

Tagline: *Knowledge should reach everyone. We build the bridge.*

2. Goals, Non-Goals and Success Measures

Goals

1. A visitor understands what Gyaansetu is, who it helps and how the app and extension fit together within about 60 seconds of scrolling.
2. Two clear, working paths: **Open the app** and **Get the extension**.
3. A memorable, polished first impression: navy blue and gold, an interactive background, an interactive cursor and smooth scroll animation.
4. The site itself is accessible (WCAG 2.2 AA). An accessibility product with an inaccessible site is a credibility failure.
5. Honest claims: only describe features that are shipped, and label the rest.

Non-goals (v1)

- Accounts, sign-up forms or any data collection on this site (that is the app's job)
- A blog, pricing, multi-page navigation or a CMS
- Analytics, cookies or tracking
- Languages other than English, apart from the Hindi brand name and short Hindi snippets

Success measures (targets)

- Both link buttons reachable by keyboard alone, and each opens its destination in one action.
- Lighthouse on mobile: Accessibility, Best Practices and SEO at least 95; Performance at least 85.
- Zero serious or critical axe violations.
- Initial JavaScript under 200 KB gzipped; no layout shift.
- A first-time reader can answer "what is it, who is it for, where do I start?" after one scroll-through (test with five people).

3. Page Flow (scroll order)

```
+-----------------------------------------------------------+
| HEADER  logo | About | App | Extension | [Open the app]   |  sticky
+-----------------------------------------------------------+
| 1. HERO          logo + tagline + scroll cue              |  #top
| 2. ABOUT US      mission, problem, who we help, values    |  #about
| 3. OUR APP       what the web app does + link             |  #app
| 4. OUR EXTENSION what it does, shortcuts + link           |  #extension
| FOOTER           disclaimer, links                        |
+-----------------------------------------------------------+
```

The order is fixed: **Hero, then About Us, then Our App, then Our Extension.** Nav anchors are `#about`, `#app` and `#extension`. The nav highlights the section in view (IntersectionObserver) and a thin gold progress line along the header shows scroll position.

4. Section Requirements

4.1 Header (sticky)

- Logo mark plus the wordmark Gyaansetu, with ज्ञानसेतु in small type.
- Anchor links: About Us, Our App, Our Extension. A gold pill button **Open the app** on the right.
- Shrinks and gains a blurred navy backdrop after the first scroll. On mobile, a menu button with `aria-expanded`; it closes on Esc and after choosing a link.
- **Accept:** every item is reachable and operable by keyboard; the current section is marked with `aria-current`.

4.2 Hero (logo and tagline)

- **Logo, large and central:** an arch bridge inside a circle in gold on navy, with the wordmark Gyaansetu and ज्ञानसेतु beneath. On first load the arch draws itself (SVG stroke animation, about 1.2 seconds), then the wordmark fades in.
- **Tagline:** *Knowledge should reach everyone.* on the first line and *We build the bridge.* on the second, in gold, with a word-by-word reveal.
- **Supporting line (one sentence):** *Gyaansetu reshapes YouTube, Physics Wallah and Unacademy lectures around how you see, hear, read and move.*

- **Scroll cue:** a small animated chevron labelled “Scroll to learn more” that jumps to About Us when activated.
- Two secondary buttons are allowed under the tagline, **Open the app** and **Get the extension**, but the hero stays visually dominated by the logo and tagline.
- Background: the interactive particle field (section 7) behind everything.
- **Accept:** the logo and tagline are visible without scrolling on a 360×640 phone and a 1366×768 laptop; the intro can be skipped by any key or click and is skipped entirely under reduced motion.

4.3 About Us (after the first scroll)

Content blocks, in order:

1. **Mission (two short paragraphs).** Draft: *Gyaansetu means a bridge of knowledge. We believe a lecture should reach every student, however they see, hear, read or move. Millions of Indian students prepare for JEE, NEET, UPSC and board exams through video lectures that were not built for everyone. We build the bridge.*
2. **The problem (three cards):** whiteboards that a screen reader cannot read; captions that are missing or weak; cluttered pages and tiny controls.
3. **Who we help (four cards):** blind or low vision; deaf or hard of hearing; dyslexia and ADHD; motor impairment. Each has an icon, one line on the barrier and one line on what Gyaansetu does.
4. **Our values (three items):** Inclusive by design; Privacy first; Built with students.
5. **Team:** cards from an editable array (photo placeholder, name, role, optional GitHub or LinkedIn link).
6. **Hackathon note (optional line):** which event this was built for, and sponsors as text only.

Interaction: blocks reveal with a staggered fade and slide as they enter; cards have a soft cursor spotlight. **Accept:** all text meets contrast targets; the section reads in a logical heading order (one H2 for the section, H3 for blocks).

4.4 Our App (after About Us)

- **Heading:** *The Gyaansetu app.* **Sub-heading:** *Set it up once. It follows you everywhere.*
- **What it does (3 to 4 bullets):**
  - Create your profile in about two minutes: pick the needs that apply to you.

    - Choose text size, speech speed and language (English India, Hindi, English US), and reading themes.
    - See live previews of your settings and manage connected devices, with a last-synced time.
    - Export or delete your data at any time.
- **Visual:** two or three app screens (onboarding, dashboard, preferences preview) in a framed device mock-up. Use real screenshots when available; until then, clearly marked placeholders.
- **Primary button: Open the app**, linked to the web app URL.
- **Interaction:** the card tilts slightly under the cursor (3D tilt with glare); the screens slide in on scroll.
- **Accept:** the button opens the live app; if the URL is not configured the button shows “Link coming soon” and is not a link.

4.5 Our Extension (after Our App)

- **Heading:** *The Gyaansetu extension.* **Sub-heading:** *Your lectures, reshaped.*
- **What it does:** applies your saved profile on **YouTube** (full support), **Physics Wallah** and **Unacademy** (basic support), on Chrome and Edge.
- **Capabilities, each with an honest status badge (live, beta or planned; text, not colour alone):**

Capability	Status
Read Screen: reads writing on the lecture video aloud (needs the server’s vision key)	live
Voice commands: pause, play, rewind, forward, speed, captions, fullscreen and more (Chrome and Edge)	live
Focus mode: hides sidebar, comments and chat on YouTube	live
High-contrast and yellow-on-black reading themes	live
Large control bar for easier clicking	live
Auto captions on YouTube where captions exist	live

Capability	Status
Physics Wallah and Unacademy tuning	beta
Dyslexia-friendly fonts	planned
Live captions and sound alerts	planned

- **Keyboard shortcuts (shown as keycaps): F** Focus mode, **G** Voice commands, **H** Read Screen, **Esc** stop speaking, **?** list shortcuts. Note: these single-key shortcuts can be switched off in the extension, and they do not fire while typing.
- **Trust note (one sentence):** *A lecture frame is sent to an AI provider only when you ask Read Screen to read it, and Gyaansetu does not store it.*
- **Primary button: Get the extension**, linked to the extension website URL (the store listing or the download page).
- **Optional later (P1):** an interactive "Try it" demo where on-screen keycaps F, G and H animate a mock lecture, clearly labelled as a simulation.
- **Accept:** the status badges match what is actually shipped; the button opens the live extension site; the shortcut list matches the extension.

4.6 Footer

- Links: Contact, GitHub, Privacy (each uses the link config; disabled state when unset).
- Disclaimer: *Gyaansetu is an independent accessibility tool and is not affiliated with Physics Wallah, Unacademy or YouTube.*
- The wordmark and ज्ञानसेतु again, and a small **Reduce motion** switch (stored in the browser).

5. Link Configuration

One file, `src/siteConfig.js`, is the only place to edit. Environment variables override it so links can be changed in the host's dashboard without a code change.

Key	Env variable	Used by	Notes
`app`	`VITE_APP_URL`	Our App button, header button, hero	Absolute `https://` URL

Key	Env variable	Used by	Notes
`extension`	`VITE_EXTENSION_URL`	Our Extension button, hero	Store listing or download page
`privacy`	`VITE_PRIVACY_URL`	Footer	Optional
`contact`	`VITE_CONTACT_URL`	Footer	`mailto:` or a page
`github`	`VITE_GITHUB_URL`	Footer	Optional

Rules

- A link is *ready* only if it is an absolute `https://` URL (`http://localhost` allowed in development).
- A link that is not ready renders as a disabled control with `aria-disabled=true`, the text “Link coming soon” and **no href**. Never `href=#`. A console warning names the missing keys in development.
- Ready external links open in a new tab with `rel=noopener noreferrer` and visually hidden text “(opens in a new tab)”.
- `npm run check-links` requests every set URL and prints PASS or FAIL with status and time, and NOT SET for empty ones. A build warns about unset links.

6. Visual Design: Navy Blue and Gold

Token	Value	Use
navy-950	#050B1E	Footer, deepest background
navy-900	#0A1633	Page background
navy-800	#0F2150	Cards and panels
navy-700	#17306B	Hover and raised surfaces
gold-300	#F3DE9A	Focus rings, highlights
gold-400	#E8C468	Headings, links, primary button fill
gold-500	#D4AF37	Borders and accents
ivory	#FFF6DD	Body text

Token	Value	Use
muted	#C9BFA0	Secondary text

- **Headings in gold; paragraphs in ivory** so long text stays readable. A subtle gold gradient is allowed only on very large display text, and every gradient stop must pass contrast.
- **Cards:** navy-800 with a thin gold border at about 35 percent opacity, 20 px radius, soft inner glow.
- **Buttons:** pill shape, minimum 48 px high. Primary: gold-400 fill with navy-950 text. Secondary: gold outline that fills on hover.
- **Typography:** Playfair Display for headings, Lexend for body, Nirmala UI or Noto Sans Devanagari for Hindi text (always marked `lang=hi`).
- **Layout:** 12-column grid, generous spacing, content width up to about 1200 px, fully responsive from 320 px upwards, no horizontal scrolling.
- **Logo:** gold arch bridge inside a circle on navy; keep clear space equal to the arch height; minimum size 24 px; an SVG favicon and a one-colour version for the footer.
- **Contrast targets:** body text at least 7:1, other text at least 4.5:1, UI borders and focus rings at least 3:1. An automated test computes the ratio for every colour pair used.

7. Motion and Interaction

Effect	Where	Technology	Reduced-motion behaviour
Logo draws itself, then wordmark fades in	Hero intro	SVG stroke animation	Skipped; logo shown
Word-by-word tagline reveal	Hero	CSS and Framer Motion	Text shown at once

Effect	Where	Technology	Reduced-motion behaviour
Interactive particle background: gold constellation with a few drifting glyph particles (ज्ञ, अ, क, A, a, ಅ); lines grab to the pointer, nearby dots repel, a click adds a few dots (capped)	Whole page, behind content, click-through	tsParticles	Replaced by a static gradient
Interactive cursor: native cursor kept; a gold dot plus a ring with spring lag; the ring grows over links and buttons and can show a short label	Pointer devices only	Framer Motion	Off
Smooth scrolling and scroll-triggered reveals, with staggered cards	All sections	Lenis with GSAP ScrollTrigger	Off; sections fully visible
Section transitions: each section fades and slides up, and the header progress line fills	Between sections	GSAP ScrollTrigger	Off
Magnetic buttons and card tilt with glare, cursor spotlight on cards	Buttons and cards	Framer Motion and CSS	Off

Rules

- Keep the native cursor visible; do not hide it. Disable the follower on touch devices.
- Never trap scrolling; keyboard scrolling, find-in-page and text selection must keep working. Anchor links scroll smoothly and then move focus to the section heading.
- Pause all loops when the tab is hidden. The particle layer scales itself down if the measured frame time rises above about 20 ms, and uses fewer dots on mobile.
- Animate only transform and opacity. No flashing or strobing.
- Reduced motion applies automatically (system setting) and through the footer **Reduce motion** switch. There is no pause button in the header.

8. Accessibility Requirements (WCAG 2.2 AA)

- Landmarks (header, nav, main, footer), exactly one H1, logical heading order, skip-to-content link.
- Everything operable by keyboard, with a visible 3 px gold focus ring on every interactive element; minimum 44 px targets.
- Meaningful link text, external links announced as opening in a new tab, disabled links announced as unavailable.
- Decorative SVG and the particle canvas are `aria-hidden`; no information is carried by colour alone (status badges have text).
- `lang=en` on the page and `lang=hi` on Hindi text. Hindi copy must be reviewed by a native speaker before launch.
- Section changes by anchor move focus and are announced; the optional demo has an `aria-live` text description of every effect.
- Automated axe checks plus a manual keyboard-only walk-through and a screen reader pass (NVDA) before launch.

9. Technical Plan

```
hub/
|-- public/                favicon.svg, robots.txt, 404.html
|-- src/
|   |-- siteConfig.js      links, features, shortcuts, team (the only file
to edit)
|   |-- components/        Header, Hero, About, AppSection,
ExtensionSection, Footer,
|   |                      Logo, LinkButton, Keycap, Background, Cursor
|   |-- hooks/             useReducedMotion, useActiveSection, useMagnetic
|   |-- styles/            tokens, globals
|   `-- App.jsx
|-- scripts/               check-links.js, contrast test
|-- vercel.json            security headers
`-- README.md
```

- **Code-split** GSAP, tsParticles and the optional demo so the first load stays small; lazy-load the particle layer after first paint.
- **Fonts** are self-hosted (Lexend, Playfair Display) and preloaded.
- **Hosting:** root directory `hub`, framework Vite, environment variables for the five links.

- **No** analytics, cookies, third-party scripts or tracking pixels.

10. SEO, Security and Privacy

- Title: *Gyaansetu – Accessible learning for every student*. Meta description, Open Graph and Twitter tags (text; add an image URL when one exists), theme colour #0A1633, SVG favicon, robots.txt and a 404 page.
- Security headers in `vercel.json`: `X-Content-Type-Options: nosniff`, `Referrer-Policy: strict-origin-when-cross-origin`, `X-Frame-Options: DENY` and a strict Content-Security-Policy (note any relaxation the libraries require).
- Privacy: this site collects nothing. The trust note on the extension section and the footer disclaimer must stay accurate. If the app's privacy page is live, link it.

11. Quality Gates and Test Plan

Check	Tool	Pass condition
Link logic	Vitest	Empty, invalid, valid, env override, localhost only in development
Link health	`npm run check-links`	Every set URL returns a success or redirect
Contrast	Vitest	Every used colour pair meets the targets in section 6
Accessibility	axe with Playwright	No serious or critical issues in three states: links unset, links set, reduced motion
Keyboard walk	Playwright	Menu, both link buttons, section anchors and the reduce-motion switch all work
Reduced motion	Playwright	All content visible; no particle animation; no cursor follower
Touch	Playwright	No custom cursor on touch emulation
Performance	Lighthouse (mobile)	Targets in section 2
Real devices	Manual	One low-end Android phone and one older laptop: smooth scrolling, no jank

12. Acceptance Criteria (Definition of Done)

1. Scrolling from the top shows the Hero, then About Us, then Our App, then Our Extension, in that order, with working anchor navigation.
2. The Hero shows the logo and the tagline without scrolling on common phone and laptop sizes.
3. The Our App and Our Extension buttons open the real, hosted destinations (verified with `check-links`), and show “Link coming soon” while a URL is unset.
4. Every feature, status badge and shortcut on the page matches what the extension actually does.
5. Navy and gold palette applied everywhere; contrast tests pass.
6. Interactive background, cursor, smooth scroll and section animations work on desktop, and degrade cleanly on mobile and under reduced motion.
7. Accessibility, performance and keyboard tests meet the targets above.
8. The site is deployed on a public URL and works on a phone.

13. Build Order

1. Scaffold the project, tokens, fonts and `siteConfig.js`; build LinkButton and the link logic with tests.
2. Header, Hero with the logo animation, and the About Us, Our App and Our Extension sections as static content.
3. Scroll system: Lenis with GSAP ScrollTrigger, active-section nav, progress line, section reveals.
4. Interactive background and cursor, then magnetic buttons, tilt and spotlight.
5. Reduced-motion and Reduce motion switch; accessibility pass and axe tests.
6. SEO, security headers, performance tuning, Lighthouse.
7. Deploy, add the real app and extension URLs, run `check-links`, test on real devices.

14. Risks and Mitigations

Risk	Mitigation
Heavy animation stutters on low-end phones	Lazy-load, cap particles, auto-reduce, disable on mobile if needed

Risk	Mitigation
Gold text hard to read on navy	Gold for headings only, ivory for paragraphs, contrast test in CI
Claims outrun the product	Feature statuses live in one config array and are reviewed before each release
Extension store listing not approved in time	Point the extension link at the app's download page; switch to the store link later
Dead links on launch day	Disabled "Link coming soon" state and `check-links` before every deploy
Custom cursor annoys low-vision or motor-impaired users	Native cursor stays; follower is decorative and can be turned off
Hindi copy errors	Native-speaker review before launch

15. Inputs Needed From the Team

1. The hosted web app URL and the extension website URL (or store listing).
2. Final logo artwork, or approval of the arch-bridge mark.
3. Team names, roles and links, and the hackathon name for the About Us section.
4. Screenshots of the app (onboarding, dashboard) for the Our App section.
5. A domain name, if one is planned.
6. Confirmation of which extension features are live on launch day so the badges are accurate.